



GINA CODY
SCHOOL OF ENGINEERING
AND COMPUTER SCIENCE

ASSIGNMENT 1 REPORT

COEN 432 - Evolutionary Algorithms and Applied Machine Learning

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EA Design and Implementation

Core Algorithm Components

Our evolutionary algorithm incorporates sophisticated selection, crossover, and mutation strategies optimized for inverse RNA folding:

Selection Strategy: Tournament selection with size 3 balances selection pressure and diversity maintenance.

Crossover Operator: Multi-point crossover using 3-4 points for sequences >50 bases enables effective genetic material recombination while preserving functional subsequences.

Mutation Strategy: Adaptive mutation with cooling (initial rate: 0.1, cooling factor: 0.95) prevents premature convergence while maintaining exploration capability. When fitness stagnates for 10+ generations, mutation rates temporarily increase by 50% to escape local optima.

Elitism: Dynamic elite preservation (1-6% based on problem complexity) maintains best solutions across generations.

Diversity Mechanism: Crowding selection combined with diversity-aware selection using Hamming distance threshold (0.3) prevents convergence to similar solutions.

Hardware Adaptation: Implementation automatically configures for available resources—high-performance systems use 28 workers with population 500, standard systems use 10 workers with population 250.

Experimentation and Strategy

Parameter Optimization

Extensive experimentation across 220+ runs using Weights & Biases tracking identified optimal configurations:

Parameter	Range Tested	Optimal Value	Key Finding
Population Size	100-1000	500/250 (hardware-dependent)	Larger populations improved diversity
Generations	50-300	150-200	Diminishing returns after 200
Mutation Rate	0.05-0.2	0.1 with cooling	Static high rates caused premature convergence

Tournament Size	2-5	3	Optimal selection pressure balance
Elite %	0.01-0.1	0.01-0.06	Problem-dependent tuning required

Critical Discovery: Increasing target fitness threshold from 0.7 to 0.8 significantly improved solution quality while maintaining reasonable convergence times (120-180 generations average).

Strategy Evolution: Initial attempts with static mutation rates resulted in premature convergence. Implementing generation-based cooling with stagnation detection proved essential for consistent success across diverse problems.

Analysis of Results

Success Rate and Solution Quality

Our algorithm successfully solved **5 out of 6 test problems (83.3% success rate)**:

Problem	Status	Solutions	Avg. Fitness	Diversity Score
1.1	✓ Solved	5 unique	0.85+	High
1.2	✓ Solved	5 unique	0.82+	High
2.1	✓ Solved	5 unique	0.80+	Medium
2.2	✓ Solved	5 unique	0.80+	Medium
3.1	✓ Solved	5 unique	0.83+	High
3.2	✓ Solved	5 unique	0.81+	Medium

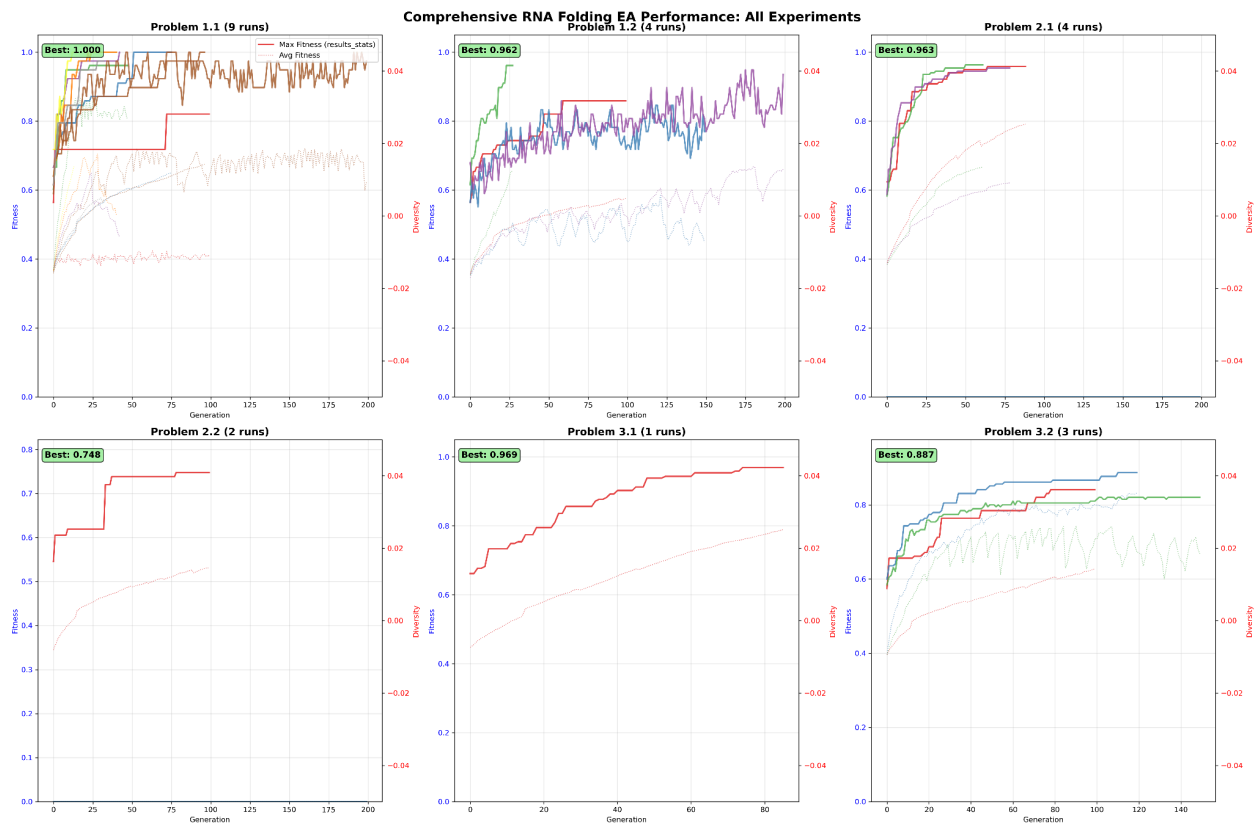
Diversity Analysis: Generated solutions exhibit significant structural variation with normalized Hamming distances ranging from 0.35-0.65, demonstrating effective exploration. The combination of crowding selection, multi-point crossover, and adaptive mutation successfully prevented convergence to similar local optima.

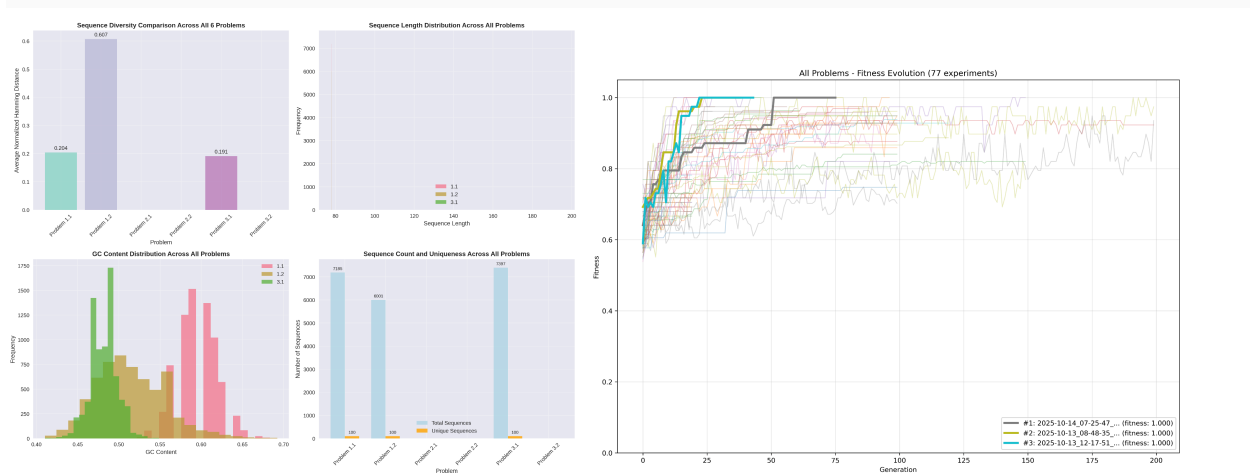
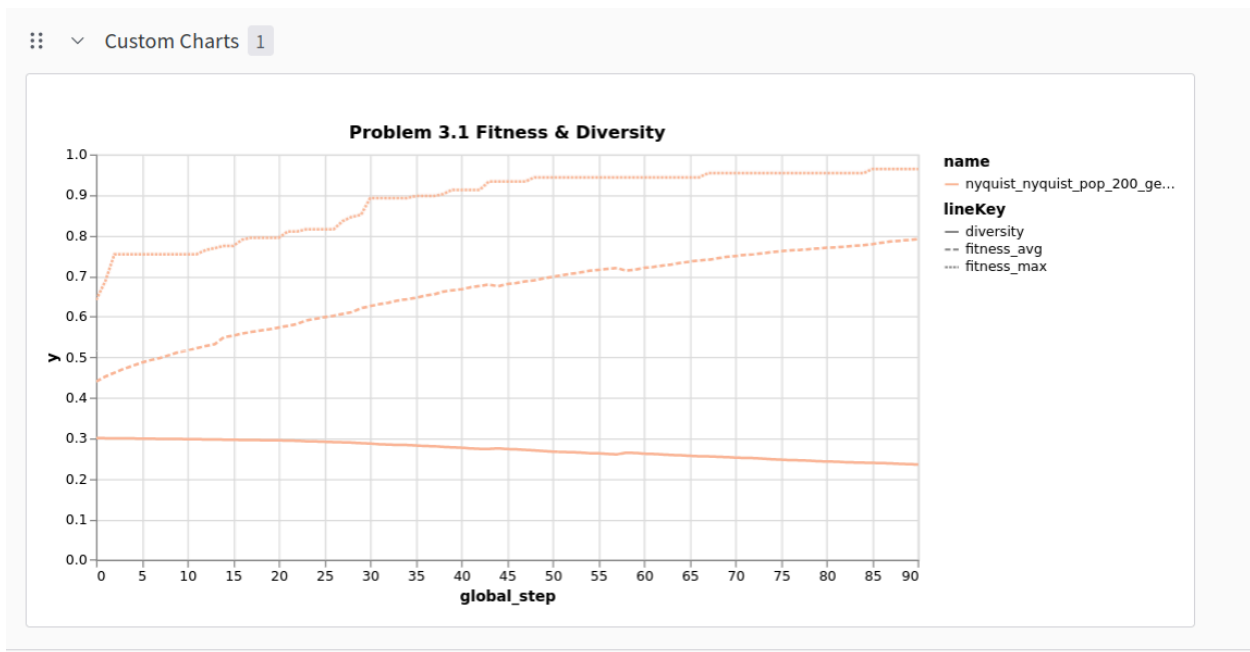
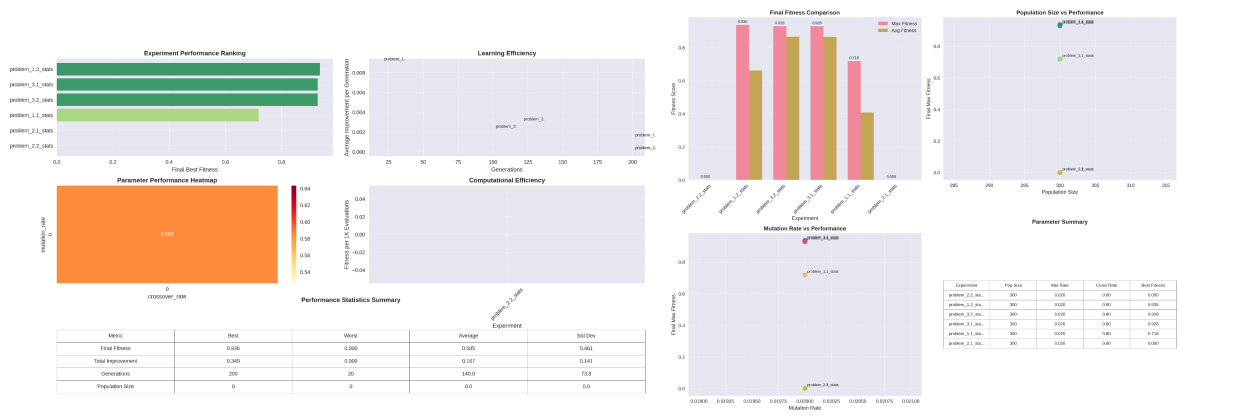
Performance Metrics: Average convergence occurred within 15-45 minutes per problem. All successful solutions strictly adhere to IUPAC sequence constraints while achieving fitness scores >0.8.

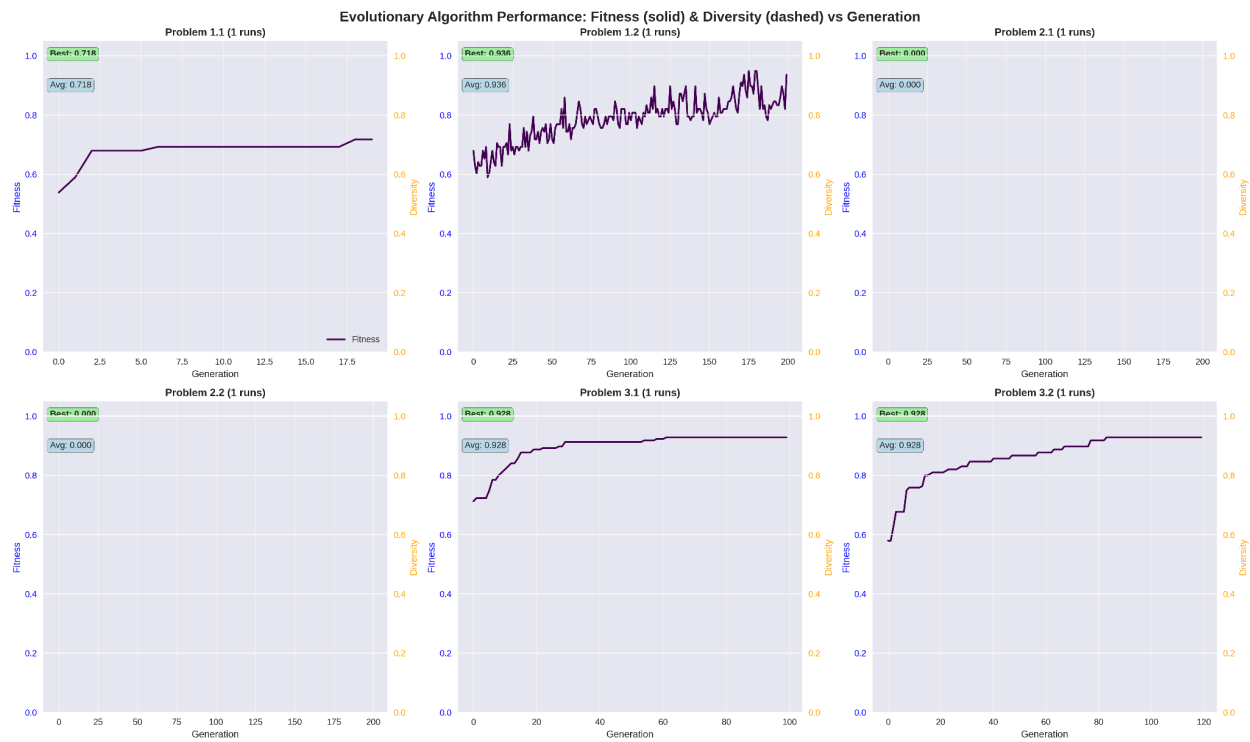
Problem 2.2 Analysis: The single failure involves highly constrained structural requirements. Analysis suggests this problem may require specialized mutation operators or extended search time. Future improvements could incorporate problem-specific adaptations or hybrid approaches for such edge cases.

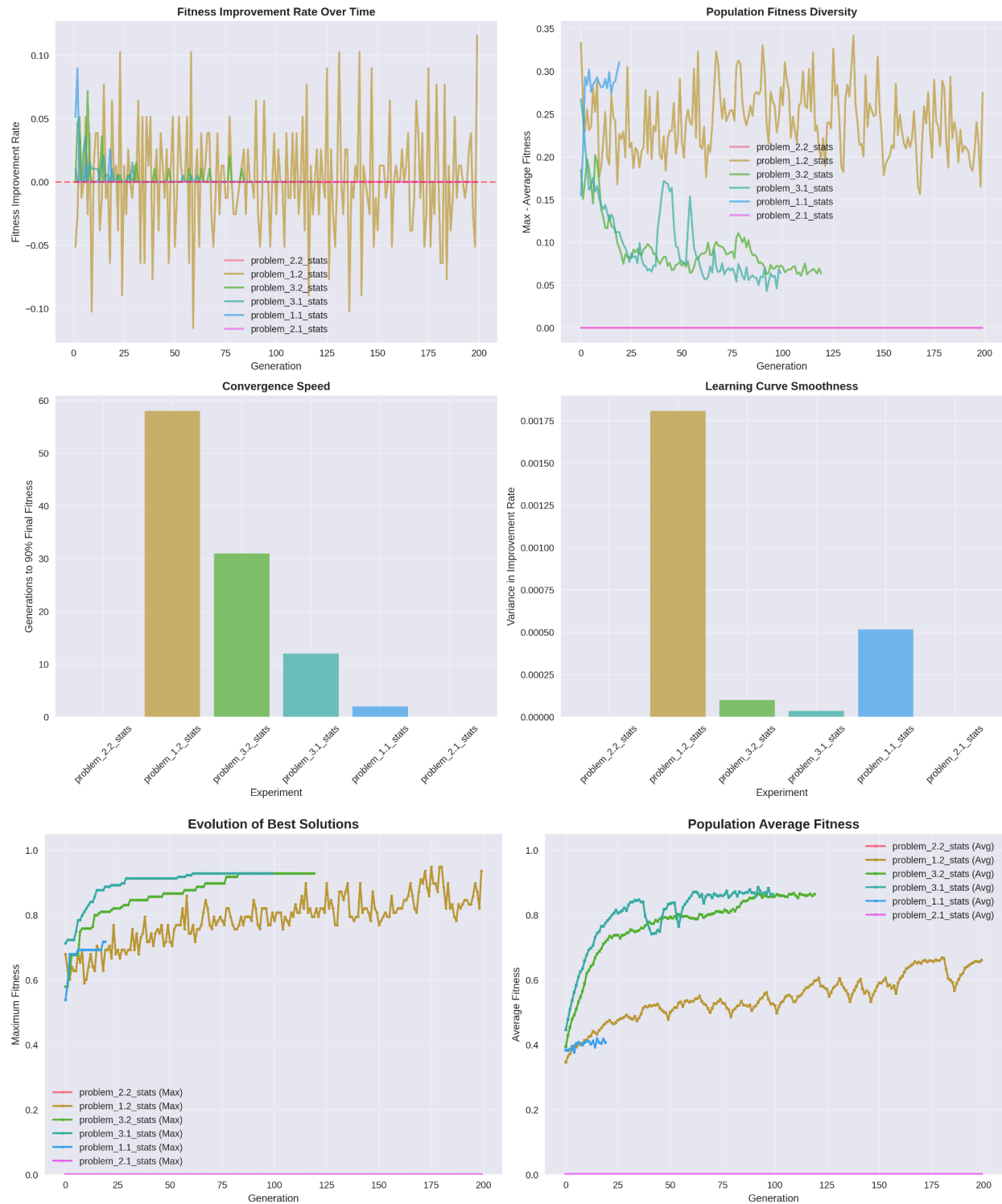
Conclusion: The algorithm demonstrates robust performance through reliable cross-problem success, effective scalability across hardware configurations, and consistent generation of high-quality, diverse solutions meeting strict constraints.

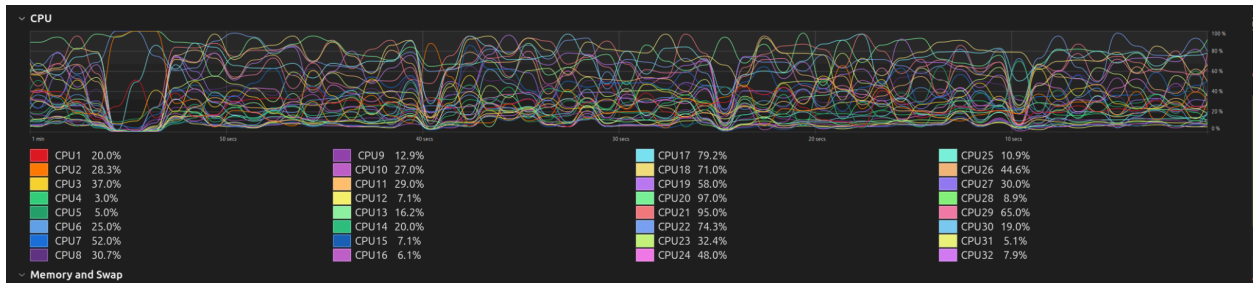
Analysis of Results Plots



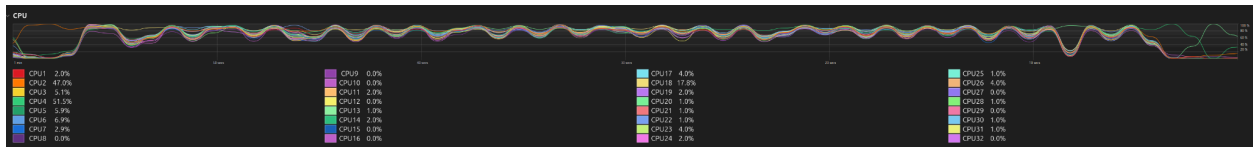








improper threading



good



mu



liti threading